

Therapeutic AAV Gene Transfer to the Nervous System: A Clinical Reality

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Gene transfer has long been a compelling yet elusive therapeutic modality. First mainly considered for rare inherited disorders, gene therapy may open treatment opportunities for more challenging and complex diseases such as Alzheimer's or Parkinson's disease. Today, examples of striking clinical proof of concept, the first gene therapy drugs coming onto the market, and the emergence of powerful new molecular tools have broadened the number of avenues to target neurological disorders but have also highlighted safety concerns and technology gaps. The vector of choice for many nervous system targets currently is the adeno-associated viral (AAV) vector due to its desirable safety profile and strong neuronal tropism. In aggregate, the clinical success, the preclinical potential, and the technological innovation have made therapeutic AAV drug development a reality, particularly for nervous system disorders. Here, we discuss the rationale, opportunities, limitations, and progress in clinical AAV gene therapy.

Introduction

The concept of gene therapy was—to our knowledge—first explicitly articulated in 1947 (Keeler, 1947). What compelled the author Clyde Keeler mostly at the time were two aspects that differentiated genetic therapeutic paradigm from other treatment strategies: the ability to address the etiology of disease and the potential to induce a “permanent correction.” To date, these arguments remain the primary justification for pursuing gene therapy. Neurological diseases, even more than other classes of disorders, can benefit greatly from the unique advantages that gene therapy aims to provide. Targeting the root cause of disease is obvious for single-gene disorders but extends to common complex disease through a variety of genetic interventions. The permanency, or at least long-lived therapeutic effect, of a gene therapy is particularly attractive for organ systems such as peripheral and sensory organs, spinal cord, or brain, which are hard to access due to the inability of most drugs to cross physiological barriers such as the blood-brain barrier (BBB). Indeed, routes of administration (e.g., injections into the brain, cerebrospinal fluid [CSF], retina, or cochlea) that are clinically prohibitive for frequent injections of a traditional drug product with short acting half-life can be now considered more easily if only a single, long-lasting intervention is required.

In the 1970s, efforts in the field to develop genetic forms of medicine became more articulated; however, we had to wait for success in clinical gene therapy for many decades to come, particularly for *in vivo* gene therapy (Friedmann, 1992). By definition, *in vivo* gene therapy relies on the direct administration of the genetic drug to the patient as opposed to *ex vivo* approaches of therapeutic gene transfer that provide a more controlled laboratory environment for the gene transfer step.

The progress in gene therapy was in lockstep with the increasing sophistication in recombinant techniques, human disease genetics, disease modeling, and molecular virology. These have now culminated in several powerful examples of clinical proof of concept such as inherited retinal disorders (IRDs) (Maguire et al., 2008) and spinal muscular atrophy (SMA) (Mendell et al., 2017).

The field of *ex vivo* gene therapy, in which tissues or cells are genetically modified in a clean room setting before they are administered to the patient, has advanced somewhat faster due to lessened safety concerns, at least in an autologous setting. In neurological disease, *ex vivo* gene therapy has previously led to demonstrable clinical benefit, e.g., in hematopoietic stem cell modification in several forms of leukodystrophy (Biffi et al., 2013; Cartier et al., 2009; Eichler et al., 2017). However, the generalizability of these approaches is limited, particularly for disease states due to cell-autonomous defects, by our current inability to transplant and functionally graft many cell types and tissues of the nervous system. The present review focuses on the clinical progress and development of *in vivo* therapeutic gene transfer.

Delivery is the key property that makes a therapy a gene therapy. The successful development of a drug relies on (1) a target, (2) a disease-modifying intervention, and (3) a delivery mechanism of said intervention to said target. While various types of interventions can be gene targeted, it is the *delivery* component that defines it as a gene therapy namely in that the drug substance itself is wholly or partly composed of genetic material. Formulating DNA or RNA to accomplish functional delivery to a cell, to a tissue, or into a complex physiological environment such as the CSF or bloodstream almost always requires a vector



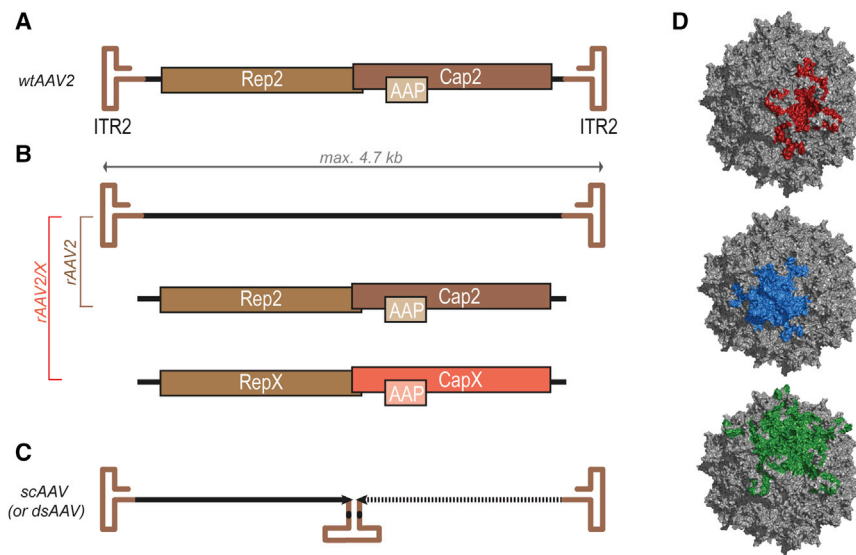


Figure 1. AAV Virus and Vector

(A) Genomic structure of the wild-type adeno-associated virus (AAV), indicating the AAV2 genes encoding the replication (Rep), capsid (Cap), and assembly-associated protein (AAP) products, flanked by inverted terminal repeat (ITR) secondary DNA hairpin structure.

(B) A recombinant AAV2-based vector is generated from the wild-type AAV2 genome by eliminating the open reading frames, solely retaining the flanking ITR sequences and replacing it with a transgene of interest up to a combined maximal sequence length of 4.7 kb. These AAV2-based vector genomes (v.g.) are packaged in the protective capsid shell by co-expressing *in trans* the AAV gene products Rep, Cap, and AAP (in addition to essential gene products from helper virus such as human adenovirus type 5). Any AAV2-based vector genome can be packaged into various different types of AAV variant capsid (CapX) by cross-packaging in which the CapX gene is co-expressed with Rep from AAV2.

(C) The genomic structure of self-complementary (sc) AAV has within the same total packaging size of 4.7 kb two complementary copies of the same transgene linked by a third ITR sequence that was modified not to be resolved during viral amplification (black dot).

(D) The AAV virion is a multimeric protein composed of 60 subunits in a 20-faceted structure with an outer diameter of approximately 25 nm. All subunits are symmetrically integrated in the particle as dimers (red), trimers (blue), and pentamers (green) according to icosahedral symmetries.

system. Indeed, unprotected (or naked) DNA or RNA injections have generally been limited in efficiency or actively rejected by the host. DNA and RNA degrading enzymatic activity, intrinsic stability, the inability to traverse lipid bilayers necessary to enter the cell and/or cellular nucleus, and the sensing mechanisms of the innate immune system are only some of the hostile elements that make *in vivo* gene delivery so challenging. While chemical modification, particularly of smaller polynucleotide chains, can provide some relief or protection to these attacks (Yin et al., 2014), formulation into a vector is most often needed to efficiently target a tissue at meaningful levels. Historically, non-viral and viral systems have been, and continue to be, explored. While each of the systems considered have unique benefits, for *in vivo* approaches that aim to provide long-lived to permanent therapeutic relief, adeno-associated virus (AAV) has emerged as a platform of increasing validation (Hastie and Samulski, 2015). Here, we discuss in detail the growing field of AAV gene transfer for neurological disorders.

AAV: Virus and Vector

AAV gene transfer is leveraging the properties of the AAV in the context of a replication-defective vector. As a wild-type virus, it is a member of the Parvoviridae family, characterized by its non-enveloped icosahedral capsid that carries a 4.7 kb single-stranded (ss) genome (Figure 1). AAVs were first identified as contaminants of laboratory adenovirus preparations. AAVs require “help” to complete a replication cycle, which can be provided by a co-infection with adenovirus, as well as a herpes virus and cellular stress. While adenoviruses can play a role in AAV’s life cycle, they are structurally and functionally highly distinct, both as a virus and in their utility as a gene transfer vector. The genome of AAV is flanked by inverted terminal repeats (ITRs), and three gene families are encoded: replication genes (Rep), structural capsid genes (Cap or VP1, -2, and -3), and a small viral

co-factor required for assembly called assembly-activating protein (AAP) (Figure 1A) (Knipe and Howley, 2013).

AAVs are considered endemic in humans and have been isolated from several other species including the non-human primate (NHP), cow, goat, mouse, and even birds (Gao et al., 2005). The route of infection remains speculative; however, in humans and non-human primates, proviral AAV DNA was detected in liver and spleen, as well as a host of other tissues, suggestive of a stage of systemic viremia and indicative of its promiscuous tropism (Gao et al., 2003). In the absence of help, AAV genomes can reside in a latent state in the host cell. How this latency is maintained remains poorly understood; however, observationally, some of the genomes reside in an episomal state as concatamers (Duan et al., 1998). In careful *in vitro* studies, infection of a wild-type virus could lead to integration into the human 19q3 locus called AAVS1 (Kotin et al., 1990; Linden et al., 1996). This relative specificity is due to the sequence homology in the ITR region and integration in the host genome dependent on the presence of Rep gene products (Linden et al., 1996). Interestingly, a natural infection with AAV is not known to lead to disease, although some reports describe an association with hepatocellular carcinoma, which remains controversial (Nault et al., 2015, 2016; Srivastava and Carter, 2017).

Recombinant AAVs are generated by elimination of all open reading frames (ORFs) from the viral genome to render it into a replication-defective virus or vector (Figure 1B). The minimal coding that needs to be retained *in cis* to allow for an AAV-like genome to be packaged is restricted to the 145 bp within the ITRs that flank the transgene within AAV vectors (Hastie and Samulski, 2015). The absence of viral genes in the vector dramatically reduces safety concerns and frees up genetic cargo capacity for the insertion of a therapeutic transgene cassette. Indeed, the overall capacity of AAV to package an ITR-flanked

genome productively is the approximate size of the wild-type AAV genome (i.e., 4.7 kb) (Figure 1B). Attempts to package larger genomes consistently lead to low titers and genome fragmentation (Dong et al., 2010; Lai et al., 2010; Wu et al., 2010). Due to its deficiency in autonomous replication, an AAV vector is produced in systems that complement the necessary components and co-factors for virion assembly, packaging, and amplification. Various systems have been developed to achieve this, particularly to support viral vector production at scale. The most common vector production is done through HEK293 transient transfection with multiple plasmids, namely one including the ITR-flanked *cis* transgene and others that encode the AAV rep and cap genes as well as a minimal set of adenoviral genes that provide help for the helper-dependent AAV to undergo viral replication (Figure 1B).

The desirable safety features of AAV as a virus and even more so as a replication-defective vector brought the system to the forefront. Decades in development, to date, AAV has been used in more than 200 human studies in which it was overall well tolerated in thousands of patients, further validating the favorable clinical profile of the platform (Ginn et al., 2018). The long development path of AAV, however, required technological advances to attain clinical success.

AAV Serotypes and the Race for Superior Capsids

Any gene delivery vector, when administered to the host, faces a hostile environment. The AAV capsid shell serves as a protection from nucleases and innate immune sensing inside and outside of the cell but also directs the tropism of the virion. Data indicate a determining role for the AAV capsid in its host response, cell attachment, cell entry, escape from the endosome, nuclear trafficking, release from the genome in the nucleus, and even its transcriptional state. Combined, the summation of these steps ultimately determines the safety and efficiency of an AAV administration following a particular route of administration (Pillay and Carette, 2017).

Variation in the primary sequence of the AAV capsid protein can alter its transduction efficiency dramatically. The first of those observations were made when exploring natural-occurring AAV serotypes. AAV1 to AAV6 were discovered in the 1960s and 1970s as contaminants of adenoviral preparations or viral cultures from tissue samples (Gao et al., 2005). These AAVs distinguished themselves serologically, hence the name serotype. AAV2, the first AAV to be molecularly cloned and rendered into a replication-defective vector, has been most extensively characterized for gene transfer applications. Other AAV serotypes were developed later as vectors, most often using a system called cross-packaging, which refers to the ability to package genomes flanked by AAV2 ITRs in any (non-AAV2) capsid by co-expressing it with the Rep genes from AAV2 during production (Figure 1B).

Early studies indicated that different AAV serotypes had a vastly distinct set of biological properties, including cell and tissue tropism (Auricchio et al., 2001). These observations sparked and continued to feed a buoying field of AAV capsid discovery and engineering. The goals of these efforts are to optimize AAV's gene transfer functionality for particular uses (e.g., improve the ability to cross the BBB or minimize immunogenicity

or to introduce novel functions (e.g., allow transduction of a previously non-permissive cell type) (Grimm and Büning, 2017).

Methodologically, these discovery approaches can be broadly categorized in three different categories: (1) bio-mining, (2) rational design, and (3) bio-panning (also referred to as directed evolution). These methods have been reviewed extensively previously, however briefly: biomining seeks to rely on nature for novel capsid diversity, rational design aims to leverage structural and functional data on capsid domains to manipulate its utility, and in bio-panning, a laboratory-based library of AAV variants is subjected to a selective pressure to then be down-selected often in an iterative sequence (not unlike phage-display methods) (Deverman et al., 2018; Grimm and Büning, 2017; Vandenberghe et al., 2009). Importantly, these methods can be combined, e.g., in recent efforts to limit the complexity of a bio-panning library within particular functional domains, e.g., based on structural or evolutionary data (Deverman et al., 2018; Grimm and Büning, 2017; Grimm and Zolotukhin, 2015; Tse et al., 2017; Vandenberghe et al., 2009).

The Vector Genome as an Engine for Optimal Transgene Expression

With all attention on the viral capsid, the critical contribution of the vector genome cannot be forgotten. The minimal component that is required in an AAV vector genome are two ITRs that flank the transgene. Minimal ITRs have a docking site for the viral Rep protein that drive the vector genome into a preformed capsid particle. ITRs are not known to have strong intrinsic promoter activity; however, recent work has identified functional transcription factor binding sites that could act as enhancer elements, at least in liver (Logan et al., 2017). The genomic cargo of a vector transgene can be designed to carry any DNA, including a cDNA (gene augmentation or addition), a silencing RNA (gene silencing) (Borel et al., 2014), other RNA species (e.g., guide RNA, long non-coding RNA, or microRNA [miRNA]-binding sites) (Han et al., 2018; Karali et al., 2011; Platt et al., 2014), anti-sense oligonucleotides (Garanto et al., 2016), a DNA- or RNA-editing enzyme (gene editing) (Senís et al., 2014), a donor template for homologous repair (Wang et al., 2015), or docking sites for DNA-binding proteins such as transcription factors (Botta et al., 2016). When expression is desired, additional regulatory sequences, such as enhancer elements and promoters (including RNA Pol II and Pol III promoters), possible intronic sequence, and terminators such as polyadenylation signals, are required (Borel et al., 2014; Deverman et al., 2018). As noted before, all these elements should ideally not exceed 4.7 kb, the genome size of the wild-type AAV (Figure 1B).

For AAV, a ssDNA virus, to become transcriptionally active, its genome needs to undergo second-strand conversion, a key rate-limiting step in the transduction pathway (Fisher et al., 1996). This bottleneck was overcome by cleverly mutating one of the ITRs in which flanking genomes in the replication cycle are resolved into two complementary separate viral or vector genomes. The net effect of this manipulation is that a single capsid is able to package a single-stranded genome with two complementing transgene sequences, connected by a (mutated) ITR hinge. These double-stranded (ds) or self-complementary (sc) AAVs can only carry a little less than 50% of the normal AAV capacity (Figure 1C) (McCarty, 2008); however, in return, scAAVs

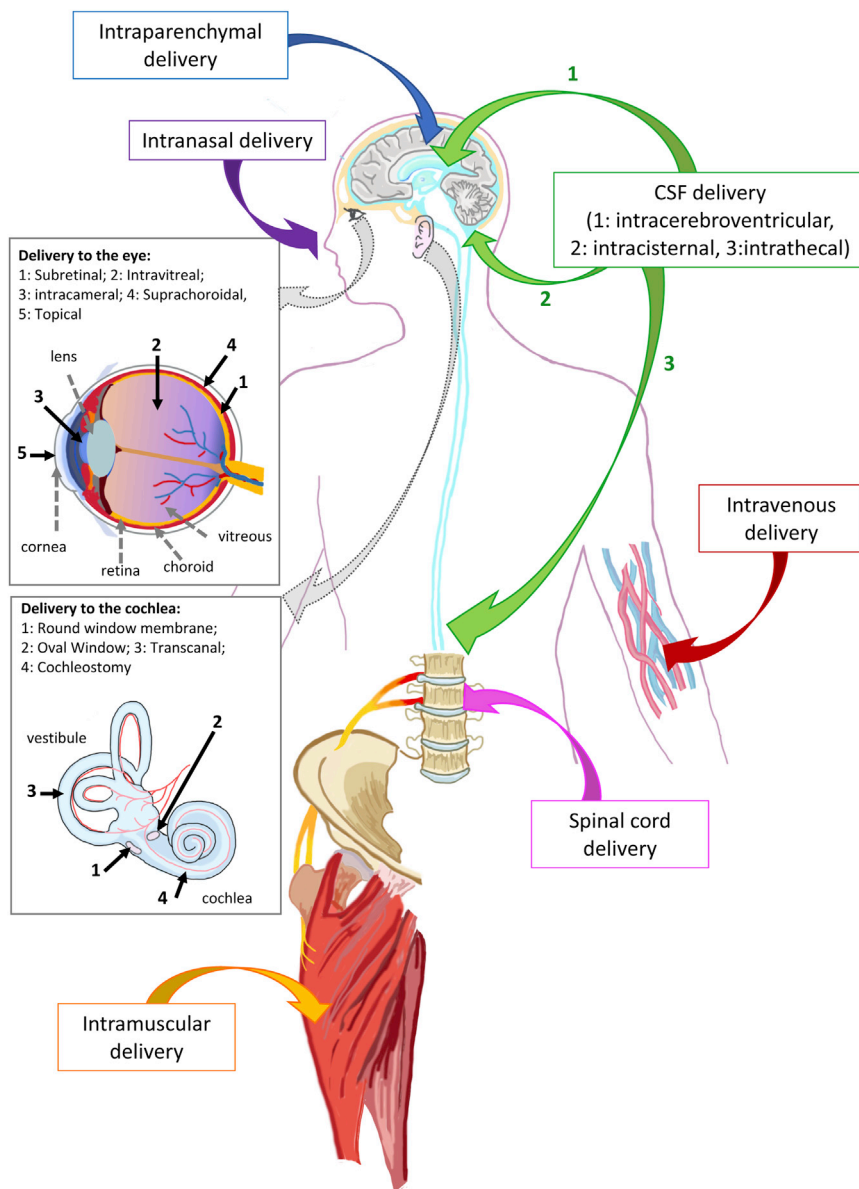


Figure 2. In Vivo Routes of AAV Delivery to the Nervous System

Local injections of vector to the eye or to the cochlea are preferably chosen for the treatment of neurosensory disorders, as a relatively small area needs to be targeted. Intraparenchymal infusion of AAV remains the most commonly used approach to deliver a therapeutic gene to the brain, even though other delivery strategies to the CSF (intracerebroventricular, intracisternal, and intrathecal routes) or to the bloodstream could be advantageous for the treatment of multifocal diseases. Intranasal delivery is an alternative and non-invasive option potentially suitable to restore the levels of therapeutic lysosomal enzymes, which are secreted and can diffuse through the CNS. Finally, a few motor neuron diseases can be improved after intramuscular or intra-spinal AAV injection.

able safety profile balanced with the appropriate level of cellular efficiency is highly dependent on the right combination of vector and route of delivery (Figure 2). The brain, eye, cochlea, and spinal cord, for example, are compartmentalized organs with physiological hurdles like the BBB greatly limiting access. Figure 2 provides an overview of the various routes of administration that are considered to target nervous system disorders in gene therapy. A subset of clinical studies using these various injection routes referred to in this section are summarized in Table 1.

Local Routes of Administration: Intraparenchymal CNS, Ocular, and Cochlear

Local routes of vector administration have clear advantages over injections into the venous system or other fluid-filled compartments. It maximizes concentration and residence time of the gene transfer agent in close proximity to the target cells, avoids wide biodistribution, and thus limits the risk of immunogenicity or

toxicities due to AAV components or ectopic expression of the transgene.

Neurosensory organs are particularly suited for local AAV delivery. Local injection to target ophthalmic disorders has progressed clinically the furthest, mainly driven by the fact that surgical access is relatively feasible, therapeutic interventions and their outcome can be monitored non-invasively, and the compartmentalized nature of the eye limits the dose and restricts systemic spread of the vector. Moreover, the retina and other targets in the eye have well-established mechanisms that promote immune privilege and thereby reduce the risk for inflammatory responses to the viral vector and transgene product. Various injection routes to administer to the eye are available and depend on the target cell type (Figure 2). Least invasive would be a topical administration on the cornea; however, AAV is only able

have been shown in multiple tissue targets to lead to a faster onset of expression and overall significantly higher sustained levels of expression (McCarty, 2008). Particularly for hard-to-access tissues such as, for example, the central nervous system (CNS) by systemic injection, the advantage of scAAV can be critical in overcoming what appears a threshold-like effect for regular ssAAVs. It is generally thought that the success of the scAAV9 in clinical trials for SMA, as discussed later, is due to AAV9 enabling BBB crossing with scAAV technology providing it the local activity upon arrival (Mendell et al., 2017).

The Route of Administration: A Major Variable Determining the Safety and Efficacy

Gene transfer to the CNS or compartmentalized sensory organs, such as the eye and the ear, is challenging. Attaining an accept-

to penetrate the tight corneal barriers inefficiently. Intracameral injections provide access to the anterior chamber and may be of interest to target the trabecular meshwork or the innervated cornea (Wang et al., 2017). Clinically, AAV programs have largely focused on degenerative retinal disorders, which are most efficiently targeted to the outer retina via subretinal injection and the inner retina and other cell types flanking the vitreous via intravitreal injection (Figure 2). In a subretinal injection, the vector is deposited between the outer retinal pigmented epithelial (RPE) cells and the neural retina. The injection of the vector bolus generates an iatrogenic retinal detachment referred to as a bleb, which resolves in animals and humans in a matter of hours. Via this route, most AAVs, within the bleb area, transduce the RPE and photoreceptor cells, a primary target for IRDs. Serotype difference influences the relative levels of transduction between RPEs and rod and cone photoreceptors, as well as limits transduction of other retinal target cells such as bipolar cells (Vandenbergh and Auricchio, 2012). AAV2 predominantly targets the RPE, a reason for its use, in Luxterna, the first FDA-approved AAV-based drug for an IRD with an enzyme dysfunction in the RPE (Russell et al., 2017). The efficiency of gene transfer following subretinal infusion is unparalleled; however, the injection procedure remains fairly complex and not routinely done clinically outside of cell and gene therapy. Some groups propose a suprachoroidal injection to provide access to the outer retina in an arguably less invasive manner (Peden et al., 2011) (Figure 2). The intravitreal injection is an attractive route that has clinically been pursued for targeting outer retinal cells, such as retinal ganglion cells, or the delivery of secreted therapeutic proteins (such as anti-VEGF agents) (Lukason et al., 2011; Trapani and Auricchio, 2018). The intravitreal injection procedure is clinically routine, is non-invasive, and has the potential to distribute across the whole retina rather than the restricted area within a subretinal bleb. However, particularly in large animal models and humans, AAV gene transfer following intravitreal injection is limited, requires high vector doses, and has a significantly increased risk of inflammation compared to subretinal injections (Miller and Vandenbergh, 2018). The appeal of intravitreal AAV gene transfer has led to a continued search for approaches and technologies that can increase its potency, reach both the inner and outer retina, and overcome the immunity obstacle to broaden its clinical use (Dalkara et al., 2013; Takahashi et al., 2017; Wassmer et al., 2017).

Like the eye, the ear presents an opportunity to deliver a vector bolus in the proximity of the neural target tissue in the cochlea at high concentration and minimal biodistribution. No clinical AAV programs have been pursued to date, in part due to the lack of precedent for intra-cochlear drug delivery; however, experimentally, Figure 2 describes the various routes that have been explored for AAV gene transfer preclinical studies (Holt and Vandenbergh, 2012; Tao et al., 2018; Yoshimura et al., 2018). In the only cochlear gene transfer study to date that has been performed in humans (NCT02132130), human adenoviral vector was injected via an oval window procedure.

Local delivery to the CNS involves a surgical procedure during which a subject is being anesthetized and restrained within a stereotaxic frame while a needle tip or a flexible fused silica catheter is directly inserted in the parenchyma through burr holes drilled

in the skull (Figure 2). For direct AAV injection in small animal models (mice and rats), stereotaxic coordinates are generally calculated from a reference point on the skull (bregma) accordingly to a brain atlas, while real-time MRI guidance systems are used for increased accuracy in larger species such as NHPs and humans (Eberling et al., 2008; Fiandaca et al., 2009; Hadaczek et al., 2006). Most AAV serotypes generally show excellent neuronal tropism (Cearley and Wolfe, 2007), while AAV5 also efficiently targets astrocytes (Davidson et al., 2000). By contrast, transduction of microglial cells appears much more challenging, with only one report so far achieving this “tour de force” using a capsid-modified AAV6 (Rosario et al., 2016). AAV intraparenchymal injections have been successful in numerous pre-clinical studies using mouse models of lysosomal storage diseases (LSDs), Alzheimer’s (AD), Parkinson’s (PD), and Huntington’s (HD) diseases (detailed in two excellent reviews, Hocquemiller et al., 2016; Choudhury et al., 2017). In those small animal models, broad transduction is achieved with only a few microliters of vector locally injected (generally $1\text{--}2 \times 10^{10}$ vg per site). Safe and widespread correction has also been reported in larger species, including feline models of GM1 and GM2 gangliosidosis (using AAV1 or AAVrh.8; McCurdy et al., 2015), dog models of Mucopolysaccharidoses types I and IIIB (using AAV5; Ellinwood et al., 2011), and NHPs after induction of a lesion in the putamen with MPTP (1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine) to mimic Parkinsonian neurodegeneration (with sustained clinical improvement observed over 8 years after delivery of AAV2-hAADC; Hadaczek et al., 2010). For other conditions for which no large animal models are available to reproduce the clinical symptoms of human diseases, safety and transduction efficacy studies after AAV intracerebral injection have been carried on in healthy NHPs and generally well tolerated (Colle et al., 2010; McBride et al., 2011; Zerah et al., 2015; Ciron et al., 2009; Sondhi et al., 2012). This modality of vector delivery has also been the first in humans for a CNS application (Janson et al., 2002) and remains the norm in most clinical programs so far (see Table 1). For motor neuron disorders, multilevel injections in the spinal cord parenchyma have been successful to deliver therapeutic agents in amyotrophic lateral sclerosis (ALS) mouse models (Azzouz et al., 2000; Franz et al., 2009; Hardcastle et al., 2018). Translation to larger mammals and humans has been challenging due to the risky surgical intervention and the focal nature of the resulting transgene expression.

Besides the risk of viral or bacterial infection, hemorrhage, and edema inherent to any neurosurgical intervention, the main limitation associated with AAV stereotaxic injection is the restricted spread of the vector in the parenchyma, especially for therapeutic applications to diseases affecting large areas of the CNS or peripheral nervous system (PNS). Long-distance anterograde and retrograde axonal transport of certain AAV serotypes (AAV1, -6, -8, and -9) has been reported and could promote dissemination of the viral particles across anatomically connected areas of the brain (Castle et al., 2014; Löw et al., 2013; San Sebastian et al., 2013; Zingg et al., 2017). Convection-enhanced delivery (CED), which consists of the application of a certain pressure during infusion to enhance the vector diffusion, is another option (Bankiewicz et al., 2000; Hadaczek et al., 2006). Other attempts to

Table 1. Ongoing Clinical Trials Using AAV for the Treatment of CNS Diseases

Current Clinical Programs of AAV Gene Therapy in the CNS (as of October 2018)

Disease	Clinical Trial phase	Serotype	Transgene	Dose	Route of administration	ID ClinicalTrials.gov	Sponsor	References
Lysosomal storage diseases								
MPS I	Phase I (recruiting)	AAV6	ZFN for safe insertion of hIDUA	5×10^{12} vg/kg 1×10^{13} vg/kg 5×10^{13} vg/kg	Intravenous	NCT02702115	Sangamo Therapeutics	Harmatz et al., 2018
MPS II	Phase I (recruiting)	AAV6	ZFN for safe insertion of hIDS	5×10^{12} vg/kg 1×10^{13} vg/kg 5×10^{13} vg/kg	Intravenous	NTC03041324	Sangamo Therapeutics	Laoharawee et al., 2018
MPS IIIA	Phase I/II (recruiting)	AAV9	hSGSH	0.5×10^{13} vg/kg 1×10^{13} vg/kg 3×10^{13} vg/kg	Intravenous	NCT02716246	Abeona Therapeutics	N/A
	Phase II/III (recruiting)	AAVrh.10	hSGSH	7.2×10^{12} vg total over 6 sites	Intracerebral	NCT03612869	LYSOGENE	Tardieu et al., 2014
MPSIIIB	Phase I/II (active, not recruiting)	AAV5	hNAGLU	4×10^{12} vg total over 16 sites	Intracerebral	NCT03300453	UniQure Biopharma B.V.	Ellinwood et al., 2011
	Phase I/II (recruiting)	AAV9	hNAGLU	2×10^{13} vg/kg 5×10^{13} vg/kg	Intravenous	NCT03315182	Nationwide Children's Hospital	N/A
LINCL (Batten disease)	Phase I (active, not recruiting)	AAV2	hCLN2	3×10^{12} vg total over 12 sites	Intraparenchymal	NCT00151216	Weill Medical College of Cornell University	Worgall et al., 2008
	Phase I/II (active, not recruiting)	AAVrh.10	hCLN2	9.0×10^{11} vg total 2.85×10^{11} vg total over 12 sites	Intraparenchymal	NCT01161576 and NCT01414985	Weill Medical College of Cornell University	N/A
	Phase I/II (recruiting)	AAV9	hCLN3	Not disclosed ("low" and "high" doses)	Intrathecal	NCT03770572	Nationwide Children's Hospital	N/A
	Phase I/II (recruiting)	AAV9	hCLN6	1.5×10^{13} vg	Intrathecal	NCT02725580	Nationwide Children's Hospital	N/A
MLD	Phase I/II (active, not recruiting)	AAVrh.10	hARSA	1×10^{12} vg total 4×10^{12} vg total over 12 sites	Intraparenchymal	NCT01801709	Institut National de la Santé et de la Recherche Médicale, France	Colle et al., 2010; Zerah et al., 2015
Hereditary leukodystrophy								
Canavan disease	Phase 1 recruiting	AAV2	hASPA	9×10^{11} vg over six sites	Intraparenchymal	NA	National Institute of Neurological Disorders and Stroke	Leone et al., 2012

(Continued on next page)

Table 1. Continued

Current Clinical Programs of AAV Gene Therapy in the CNS (as of October 2018)

Neurotransmitter disorder

AADC deficiency	Phase I/II (active, not recruiting)	AAV2	hAADC	Not disclosed	Intraparenchymal (Putamen)	NCT01395641	National Taiwan University Hospital	Hwu et al., 2012; Chien et al., 2017
	Phase II (recruiting) - expansion of NCT01395641	AAV2	hAADC	2.37×10^{11} vg total	Intraparenchymal (putamen)	NCT02926066	National Taiwan University Hospital	
	Phase I (recruiting)	AAV2	hAADC	1.3×10^{11} vg total	Intraparenchymal (SNc and VTA)	NCT02852213	University of California, San Francisco	

Idiopathic age-related neurodegenerative diseases

Parkinson's disease	Phase I/II (active, not recruiting)	AAV2	NTN	2.4×10^{12} vg total over 4 sites sham surgery	Intraparenchymal (putamen and SNc)	NCT00985517	Sangamo Therapeutics	Marks et al., 2010
	Phase I (active, not recruiting)	AAV2	hGDNF	9×10^{10} vg total 3×10^{11} vg total 9×10^{11} vg total 3×10^{12} vg total	Intraparenchymal + CED (putamen)	NCT01621581	National Institute of Neurological Disorders and Stroke (NINDS)	Johnston et al., 2009; Su et al., 2009
	Phase I (active, not recruiting)	AAV2	hAADC	7.5×10^{11} vg total 1.5×10^{12} vg total 4.7×10^{12} vg total	Intraparenchymal (striatum)	NCT01973543	Voyager Therapeutics	Bankiewicz et al., 2006
	Phase II (recruiting)	AAV2	hAADC	2.5×10^{12} vg total versus sham	Intraparenchymal (striatum)	NCT03562494	Voyager Therapeutics	
	Phase I (active, not recruiting)	AAV2	hAADC	9.4×10^{12} vg total	Intraparenchymal (striatum)	NCT03065192	Voyager Therapeutics	
	Phase I/II (recruiting)	AAV2	hAADC	3×10^{11} vg total 9×10^{11} vg total over 4 sites	Intraparenchymal (putamen)	NCT02418598	Jichi Medical University	Christine et al., 2009
Alzheimer's disease	Phase I	AAV2	hNGF	2.0×10^{10} vg total 1.0×10^{11} vg total 2.0×10^{11} vg total	Intraparenchymal (basal forebrain region)	NCT00087789	Ceregene	Rafii et al., 2014, 2018
	Phase I (not yet recruiting)	AAVrh.10	hAPOE2	8.0×10^{10} gc/kg 2.5×10^{11} gc/kg $8.0 \times 10^{-}$ gc/kg	Intracisternal	NCT03634007	Weill Medical College of Cornell University	Rosenberg et al., 2018

(Continued on next page)

Table 1. Continued

Current Clinical Programs of AAV Gene Therapy in the CNS (as of October 2018)

Spinal cord disorders					
SMA1	Phase I (recruiting)	AAV9	SMN	6.0 × 10 ¹³ vg total 1.2 × 10 ¹⁴ vg total	Intrathecal
					NCT03381729
	Phase III (recruiting)	AAV9	SMN	1.1 × 10 ¹⁴ vg/kg	Intravenous
					NCT030505099
	Phase III (active, not recruiting)	AAV9	SMN	Not disclosed	Intravenous
					NCT03306277
	Phase III (recruiting)	AAV9	SMN	Not disclosed	Intravenous
					NCT03461289

Active clinical interventional AAV studies listed on <https://clinicaltrials.gov> as of October 2018 are included while completed or terminated studies are not listed. Associated references correspond either to studies demonstrating the proof of concept of a strategy in animal models other than mice or to clinical results in patients already published.

Abbreviations: MPS, Mucopolysaccharidosis; LINCL, late infantile neuronal ceroid lipofuscinosis; MLD, metachromatic leukodystrophy; IDUA, alpha-L-iduronidase; IDS, iduronate-2-sulfatase; SGSH, N-sulfoglucosamine sulfohydrolase; NAGLU, alpha-N-acetylglucosaminidase; ARSB, arylsulfatase B; CLN2, protein tripeptidyl peptidase-1; AADC, aromatic L-amino-acid decarboxylase; NTN, neurturin; GDNF, glial-cell-derived neurotrophic factor; NGF, nerve growth factor; APOE2, apolipoprotein 2; SMA1, spinal muscular atrophy type 1; SMN1, survival motor neuron 1; SNc, substantia nigra pars compacta; VTA, ventral tegmental area.

potentiate AAV transduction after local delivery in the brain include co-infusion of factors such as heparin (Mastakov et al., 2002) or mannitol (Mastakov et al., 2001), in possible combination with CED (Carty et al., 2010). Finally, in specific neurological contexts such as neuropathic LSDs, the release of therapeutic lysosomal enzymes from transduced neurons and their uptake by distal cells leads to an expansion of the corrective effect of the gene therapy to larger areas than where AAV particles diffuse. This cross-correction process depends upon the presence of mannose 6-phosphate receptors on recipient cells and is conserved among mammals. Considering that relatively low residual enzymatic activity can be clinically relevant (Leinekugel et al., 1992), intracerebral delivery of therapeutic genes for the treatment of infantile LSDs remains the main application for AAV-based gene therapy in the clinic (Table 1).

Delivery to the CSF via Intracerebroventricular, Intracisternal Delivery, or Intrathecal Routes

Delivery to the CSF allows for more widespread gene transfer throughout the brain and spinal cord (Hardcastle et al., 2018) while limiting systemic biodistribution (Figure 2). However, the presence of tight junctions between ependymal cells restricts such application to certain AAV serotypes (such as AAV1, -2, -4, -5, -8, -9, rh10, and PHP.B) able to extravagate through this protective cell layer (Hinderer et al., 2018a). Importantly, the age of the animals at the time of intervention influences the transduction profile following AAV infusion in the CSF: for example, a comparative study demonstrated that intracerebroventricular (i.c.v.) injection of AAV8 and AAV9 resulted in widespread biodistribution in the brain as compared with other serotypes (AAV1, -2, -5, and -7) when administered on postnatal day 0 (P0) and that the transduction profile tended to shift from neurons to glial cells between P0 and P1–P2 (Chakrabarty et al., 2013).

In pre-clinical studies using rodents, i.c.v. or intracisternal (IC) infusions of AAV in neonates and adult mice lead to widespread expression of transgenes across the cerebral tissue and the spinal cord, a strategy that has been successful in correcting symptoms of MPS I (AAV8-hIDUA; Wolf et al., 2011), II (AAV9.hIDS; Hinderer et al., 2016), IIIA (AAV9.CAG; Haurigot et al., 2013), VII (AAV1-GUSB; Passini et al., 2003), GM1 gangliosidosis (Broekman et al., 2007), SMA (AAV9.SMN1; Armbruster et al., 2016), MDL (AAV1-hARSA; Hironaka et al., 2015), ALS (AAV.miRNA-SOD1; Dirren et al., 2015), and even AD (Levites et al., 2006). Similar strategy has been attempted in larger species such as dogs for the treatment of ceroid lipofuscinosis (Biferi et al., 2017) and MPS IIIA (Haurigot et al., 2013) and cat models of Sandhoff disease (Rockwell et al., 2015). Interestingly, transduction of the ependymal cell layer and/or the choroid plexus by AAV2 or AAV4 also improved disease symptoms without requiring the transport of viral particles across this biological barrier (Liu et al., 2005; Hudry et al., 2013; Benraiss et al., 2013). In those cases, it is assumed that the ependymal and choroid plexus cells act as therapeutic reservoirs for the secretion of soluble enzymes or neurotrophic factors throughout the cerebral tissue. Whether or not long-term gene expression can be achieved remains unknown, especially because of the relatively fast turnover rate of the ependymal cells (Chauhan and Lewis, 1979).

By comparison, intrathecal (IT) delivery of AAV vectors at the level of the lumbar cord is easily translatable to the clinic considering that this space is routinely accessed into via lumbar puncture. Additionally, the insertion of a catheter through the lower section of the spinal cord can eventually be advanced rostrally to the cisterna magna with limited risks. Such an approach has been used for the preclinical evaluation of therapeutic candidates (serine histogranin, endomorphin1) for the relief of neuropathic chronic pain symptoms (using an AAV8; Jergova et al., 2017) and to correct ALS neurodegenerative processes in SOD1 mice (using an AAVrh.10; Li et al., 2017). A single IT injection of AAV.hGAA also led to neurologic and cardiac improvements in Pompe disease (Hordeaux et al., 2017), while similar central and peripheral benefits could be demonstrated after infusion of AAV9, AAVrh.10, and AAV.Olig001 in young Krabbe disease mice (10–11 days old; Karumuthil-Melethil et al., 2016). Overall, the transduction efficacy appears to be rather equivalent in adult mice when comparing different routes of administration in the CSF (Bey et al., 2017), but infusion of AAV9 in the cisterna magna or in the lateral ventricle has been shown to be far more potent at transducing the brain and spinal cord of cynomolgus macaques or dogs as compared with IT injection by lumbar puncture (Hinderer et al., 2018a). Whether or not placing the subject in the Trendelenburg position right after infusion can potentiate the efficacy of brain transduction by this procedure remains debated (Meyer et al., 2015).

Intranasal Delivery for Brain Transduction

While the primary indication for intranasal AAV delivery has been to target the respiratory tract, this non-invasive gene transfer strategy has also been recently applied for the treatment of MPS I (Belur et al., 2017) (Figure 2). Adult IDUA-deficient mice were instilled intranasally with an AAV9.IDUA, and a complete restoration of enzyme activity to normal levels was noticed 5 months post treatment in all parts of the brain, even though the area of AAV transduction remained localized to the olfactory bulb. This approach may be suitable for other LSDs considering the possible trafficking of therapeutic lysosomal enzymes to the CNS via the olfactory and trigeminal tracks innervating the nasal passages (Wolf et al., 2012). Intranasal delivery of an AAV vector driving expression of the neurotrophic factor BDNF fused with cell-penetrating peptides (TAT and HA2) was also associated with anti-depressant effects in mice and is considered a promising therapy for major depressive disorder (Ma et al., 2016).

Intramuscular Injection and Nerve Delivery to the Spinal Cord

Targeting the spinal cord can be achieved via remote delivery in the muscle and relies on the ability of certain AAV serotypes to hijack the axonal transport machinery and travel along the nerve following a centripetal way (Boulis et al., 2003; Towne et al., 2010) (Figure 2). AAV1, AAV6, and, more recently, AAV9 have led to efficient targeting of motor neurons after intramuscular infusion in rodents (Benkhalifa-Ziyyat et al., 2013; Hollis li et al., 2008; Towne et al., 2010). Infusion of a ssAAV6 in the gastrocnemius muscle of African green monkeys has been attempted and transduced about 50% of motor neurons in the ventral horn of the ipsilateral-injected side of the spinal cord (1.3×10^{12} vg injected; Towne et al., 2010). AAV intramuscular injections appear especially suited for the treatment of motor neuron diseases

such as ALS, SMA, or pain (Benkhalifa-Ziyyat et al., 2013; Wang et al., 2002). Unfortunately, scaling up the dose of vector to treat larger animals has proven challenging, the spread of the vector remaining limited and localized in the lumbar segment of the spinal cord. Additionally, early dysfunction of the axonal transport machinery in those diseases may prevent efficient AAV transduction (Bilsland et al., 2010; Fischer et al., 2004).

Intravenous Delivery for Non-invasive and Broad Transduction of the CNS

The BBB efficiently isolates the CNS from the peripheral compartment and is a complex biological barrier constituted by successive layers of brain endothelial cells coherently connected by tight junctions, pericytes, and astrocytic endfeet (Figure 2). Numerous efforts have been made to try to disrupt the BBB in order to allow the transfer of therapeutic agents. Infusion of mannitol, for example, has been used to potentiate the transduction of neural tissue after intravenous delivery of AAV (Mastakov et al., 2001; McCarty et al., 2009), an approach somewhat successful but associated with significant adverse effects.

Because of those challenges, the characterization by Foust and colleagues of an AAV (AAV9) able to naturally bypass this “impermeable” biological barrier constitutes a definite breakthrough in the field (Foust et al., 2009). For the first time, large-scale transduction of the entire brain and widespread expression of therapeutic genes in the CNS became achievable after a single intravenous injection of vector, a feature especially advantageous for the treatment of multifocal diseases (Foust et al., 2010; Mattar et al., 2013; Murrey et al., 2014). Indeed, illustrated in Figure 3, the combination of scAAV and AAV9 at high dose can result in widespread transduction of spinal cord and the CNS following an IV injection (Hudry et al., 2018). This finding served as the basis of many preclinical and clinical programs described in more detail below. It also spurred further technology exploration. Indeed, other AAV serotypes appear to have qualitatively similar transduction abilities (Yang et al., 2014; Zhang et al., 2011), even though those remain less well characterized across various disease and animal models than AAV9. Capsid engineering studies have further led to AAV variants that quantitatively appear superior to AAV9 such as AAV.PHP.B and AAV-B1 (Choudhury et al., 2016a; Deverman et al., 2016) and AAV-AS (Choudhury et al., 2016b), at least in the animal model tested in the studies (AAV.PHP.B appears to not sustain that superior ability in other species; Challis et al., 2019; Hordeaux et al., 2018). Of note, most of the successful attempts of systemic AAV9 delivery have been done after packaging self-complementary genomes (Figure 1C). Anc80L65, a fully rationally designed AAV, has been shown to—at least in part—overcome the need for a self-complementary genome to accomplish substantial CNS transduction following systemic injection (Deverman et al., 2016; Hudry et al., 2018; Zinn et al., 2015).

Due to its extensive validation studies across models and by multiple groups, AAV9 remains the gold standard serotype for CNS targeting after systemic delivery and has been extensively used with minimal signs of peripheral or central toxicity (Murrey et al., 2014; Plegier et al., 2011; Saraiva et al., 2016). Interestingly, notwithstanding some discrepancies between the reported studies, the transduction profile in the brain in animals has been reported to shift accordingly to the age of treatment: while

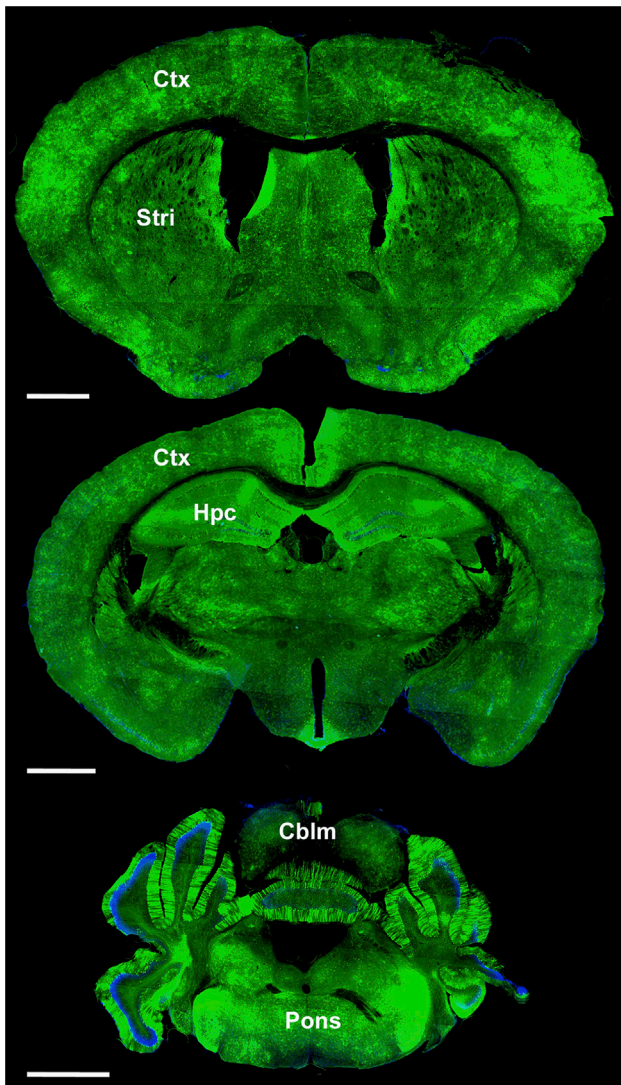


Figure 3. AAV9 Brain Transduction after Intravenous Delivery in C57BL/6

Representative images of eGFP fluorescence signal (and DAPI) across three different coronal sections across the brain after intravenous injection of AAV9 harboring a self-complementary genome (4×10^{13} vg/kg) expressing CMV.eGFP (Hudry et al., 2018) at the level of the striatum (upper panel), hippocampus (middle panel), or cerebellum (bottom panel). C57BL/6 animals were injected via tail vein at 6–8 weeks of age. Scale bar, 1,000 μ m.

neurons are preferentially targeted after delivery in neonates, astroglia mostly express the transgene after infusion in adult mice (Deverman et al., 2016; Foust et al., 2009; Hudry et al., 2016). Systemic injection of AAV9 in newborn or adult mice transduced motor neurons (MNs) in the spinal cord, up to 78% in lower spinal MNs according to Wang and colleagues in newborn mice (Wang et al., 2010) but with declining efficiencies later in life (Duque et al., 2009; Gray et al., 2011). AAV9 studies in NHPs confirmed those initial findings, showing efficient transduction of neurons after intravenous infusion in neonatal rhesus macaques, while glial targeting was prominent after administration in juvenile subjects (Gray et al., 2011; Mattar et al., 2013). MNs were consis-

tently targeted regardless of the age of the subject. Preclinical results have been rather positive so far (for GM1 gangliosidosis [Weismann et al., 2015], Sandhoff disease [Walia et al., 2015], MPS IIIB [Fu et al., 2011; Murrey et al., 2014], AD [Iwata et al., 2013], HD [Dufour et al., 2014], SMA [Benkhelifa-Ziyyat et al., 2013], or ALS [Yamashita et al., 2013]). Importantly, the first clinical report of gene-replacement therapy for SMA type I has recently been published and showed remarkable safety and efficacy of this approach (Mendell et al., 2017).

Even though AAV peripheral delivery is technically easy and non-invasive, important challenges complicating clinical application remain: (1) large doses of vector are required to achieve relevant transduction in the CNS and improve disease symptoms; (2) intravenous administration exposes the vector to anti-AAV antibodies in those individuals that were previously exposed to AAV; (3) as most AAVs are hepatotropic, most of the vector will distribute to the liver, increasing the risk for hepatotoxicity; and (4) ectopic expression of the transgene in peripheral tissues may further pose safety risks.

In summary, the choice of the route of administration is a critical one yet requires a balancing of multiple parameters along a risk-benefit equation. Historically, local routes of injection have been preferred as they minimize the safety considerations while maximizing delivery. Driven by the desire to overcome the invasiveness of the local administration procedure and/or the clinical need to target more broadly than what a local injection route can deliver, novel technologies were, and often still are, needed that enable meaningful levels of gene transfer via these routes. These approaches, at the current stage, almost always require higher levels of dosing, increasing the safety risk and putting significant burden on AAV manufacturing. In certain cases, various routes of administration are combined when either technologies are too limiting or the disease presentation (e.g., in syndromic cases) requires multiple organs to be targeted (Biferi et al., 2017; Gurda et al., 2016).

Safety Consideration in AAV Gene Therapy

A primary driver for AAV's adoption in gene therapy approaches has been its remarkable safety record in preclinical and clinical studies. Even at high doses, traditional toxicology assessment is relatively uneventful. However, one cannot be complacent as the first reports of dose-limiting toxicities using AAV emerge. Indeed, in a recent study in piglets and NHPs, an AAV variant administered i.v. at high dose led to severe toxicity that resulted in death or necessitated euthanasia days after vector administration. Observations included elevated transaminases, degeneration of dorsal root ganglia, impaired ambulation, proprioceptive deficits, and ataxia (Hinderer et al., 2018b). AAV gene therapies are complex biological agents for which the clinical experience of the platform remains in its infancy. Moreover, host responses can be unpredictable and dependent on disease state, route and compartment of administration, genetic predispositions, or other idiosyncratic variables.

Host Response

Immunological responses to a gene therapy can significantly impact the safety and longevity of any gene therapy. Possible sources of antigenicity are the vector components, the

expressed transgene, and contaminants of the vector preparation. The immunogenicity can be innate and/or adaptive in nature.

Gene therapies, like with other biologics, can be complicated by pre-existing immunity (PEI). PEI is due to prior exposure of the host to one of components in the gene therapy and the lasting memory immune responses to which the initial exposure may lead. Since AAV is endemic in humans, PEI complicates the development of gene therapies at various levels as most elements that make up the gene therapy are naturally derived and thereby possible to have pre-exposed the recipient. Individuals with memory B cell responses have circulating levels of anti-AAV neutralizing antibodies (NABs) that can, even at low levels, prevent the AAV particle from reaching its intended tissue or cell target. The most immediate impact of AAV NABs is therefore on (the lack of) efficacy, not safety. Sequential AAV injections may suffer from NABs elicited by a prior administration, and re-administration is therefore currently not considered a feasible option (Amado et al., 2010). PEI to AAV can also manifest itself through a memory T cell response; indeed, the reactivation of memory T cell responses has been hypothesized to play a role in hepatotoxicity and the loss of transgene expression that has been observed in several AAV clinical trials. The prevalence and impact of PEI to AAV are significant but difficult to gauge quantitatively due to differences in the detection assays and the vector serotypes tested between populations (e.g., age, clinical history, geographical location). General concepts on immunological and other safety considerations for AAVs have been reviewed extensively elsewhere (Mingozzi and High, 2013; Vandamme et al., 2017; Vandenberghe and Wilson, 2007).

PEI is not solely an issue of vector capsid, as it can also be directed to the expressed transgene. An example of this in a traditional gene augmentation context is LSD for which enzyme replacement therapies are available (Wang et al., 2008). During these treatments, some patients develop antibody responses to the injected enzyme. If these patients later are receiving a gene therapy expressing the same enzyme, these NABs constitute (drug-induced) PEI in the context of the gene therapy. Another example relates to the use of CRISPR genome editing, as the bacterial CRISPR enzyme itself is an antigen that humans are exposed to throughout life (via a common bacterial infections) (Charlesworth et al., 2018).

Fortunately, AAV gene therapies for neurological disease have been somewhat spared from major inflammatory adversity, likely due to the immune privilege that many of the compartments in the nervous system benefit from (Carson et al., 2006; Forrester et al., 2018). Brain, spinal cord, eye, and cochlea have limited access to circulating antibodies or infiltrating immune cells and, in certain cases, mechanisms to actively dampen immune response (Forrester et al., 2018; Stein-Streilein, 2013). NABs due to PEI, for example, are significantly reduced in the ocular vitreous or CSF. Intraparenchymal CNS or subretinal injections even have less exposure to NABs and likely are not affected, even if the host is highly serologically positive or has previously received an AAV injection (e.g., when administering a retinal gene therapy in both eyes) (Amado et al., 2010; Gray et al., 2013). Indeed, clinical programs that relied on direct intraparenchymal or intra-CSF injections overall confirmed the safety and

tolerability of these approaches (detailed in Table 1 and in the section “Clinical Progress: Illustrative Case Studies”), with little signs of inflammation reported and only a small subset of patients eventually developing a mild immune reaction against the vector capsid. Similarly, in the eye, programs using a subretinal route of injection have overall safely progressed in the clinical studies, presumably due to the local immune privilege of the retina, the small dose, and the containment of the vector in the subretinal bleb (Trapani and Auricchio, 2018). Nonetheless, immune privilege is a relative concept, as in higher animal studies inflammatory responses have been noted particularly when higher doses are administered, possibly due to innate immune sensing in the retina or leakage of virus into the vitreous due to the subretinal injection procedure (Khabou et al., 2018; Reichel et al., 2017; Vandenberghe et al., 2011). Direct intravitreal, which conceptually has noted advantages over subretinal injection, illustrates the limits of immune privilege even more. Contrary to many other commonly used protein-based drugs, AAV injected in the vitreous was found to be inflammatory, even at moderate doses (Miller and Vandenberghe, 2018). These responses remain poorly understood, mainly because they are only evident in larger animal models and human studies (Cukras et al., 2018; Maclachlan et al., 2011), and while responsive to pharmacological immune suppression, further research is needed to adequately avoid or mitigate these responses.

Systemic AAV administration with the goal to broadly transduce the neural tissue is evidently more challenging than local injection routes. Most studies are forced to exclude patients with detectable levels of NABs to AAV, as it is expected they will not benefit from the gene therapy. Even then, AAV clinical studies for hemophilia showed that seronegative subjects developed transaminase elevations weeks after AAV administration, which in some patients was concomitant with a rise in AAV capsid-specific CD8⁺ cells and the loss of transgene expression (Manno et al., 2006; Mingozzi et al., 2007; Nathwani et al., 2011). In the SMA study in which AAV9 was given IV at high dose, asymptomatic elevation of transaminases was observed in some of the newborn subjects, who fortunately responded to prednisolone treatment (Mendell et al., 2017).

Cellular immune response toward heterologous or even autologous gene products poses another concern to take into account when considering AAV gene transfer. Even after direct intraparenchymal or intra-CSF injection, a few pre-clinical studies in dog models of MPS IIIA disease, Sanfilippo, and Hurler syndromes have actually showed that a robust inflammatory response can arise from AAV-driven expression of human transgenes, while no such response was triggered when the canine cDNAs were cloned in the vector backbone (Ellinwood et al., 2011; Haurigot et al., 2013).

Ectopic expression of the transgene product can be a source of antigenicity or toxicity, an issue that warrants the development of strategies to restrict expression of the therapeutic protein to targeted tissues. Capsid modifications can be made to reduce liver targeting and promote transduction of the neural tissue (Castle et al., 2016; Kanaan et al., 2017). Another successful strategy has been to tailor the AAV genome and insert cell-specific promoters allowing specific expression in defined cell populations such as neurons (synapsin or CamKII promoters;

Jackson et al., 2016; McLean et al., 2014; Watakabe et al., 2015), astrocytes (GFAP or ALDH1/1 promoters; Dashkoff et al., 2016; Koh et al., 2017), or oligodendrocytes (using a myelin basic protein promoter; Georgiou et al., 2017; Kennedy and Rinholm, 2017). miRNA-specific target sequences have also been cloned in the viral genome (3' UTR) to abolish off-target expression in undesirable cell types (Taschenberger et al., 2017).

Currently utilized clinical strategies to side step or overcome immune concerns brought on by AAV gene therapies include the exclusion of patients with elevated anti-AAV antibodies from enrolling in clinical trials, administering in compartments that are thought to be immune privileged, including short-term immunosuppression regimens (e.g., oral prednisone; Mendell et al., 2017; Nathwani et al., 2011), or limiting the vector dose (Mendell et al., 2017). While these strategies have overall proven to be adequate in the ongoing clinical studies, it is clear they do not provide a solution for many of the outstanding concerns in the field. How can we effectively administer an AAV to a seropositive patient? Can we do repeat administrations of AAV over time? How do we safely perform AAV gene transfer with an antigen that is foreign to patient, such as when a patient has null mutations in an autosomal recessive disease gene? These and other questions remain the topic of investigation and technology development in the laboratory (reviewed in Mingozi and High, 2013).

Genotoxicity and Insertional Mutagenesis

Contrary to retroviral vectors, recombinant AAV is not known to actively integrate into the host genome but rather resides in a non-replicating episomal state upon transduction in the target cell (Schnepp et al., 2005). Indeed, since no mechanism actively duplicates the transgene during mitosis, its expression is eventually lost in a dividing cell population. However, infrequent integration of the AAV transgene into the host genome can occur, with a bias toward active genomic regions (Nakai et al., 2003). The risk for generating insertional mutagenesis therefore does exist and was recently reviewed in detail (Chandler et al., 2017). While this safety issue has not yet been reported in human patients, increased incidence of hepatocellular carcinoma (HCC) has been observed after systemic delivery of AAV2 and AAV9 in newborn MPSVII mice (AAV2) and Sandhoff disease model as the animals get older (between 43 and 52 weeks post injection) (Donsante et al., 2007; Walia et al., 2015; Zhong et al., 2013). The incidence of those events reaches 30%–60% after injection of AAV2 at a dose of 1.5×10^{11} vg and 80% with AAV9 at the much higher dose of 2.5×10^{14} vg with similar observations in non-transgenic littermates. Further investigation of the genetic material of the tumors revealed the presence of recombinant AAV genome integrated in a specific locus of the murine chromosome 12 (*Rian* locus), leading to dysregulation of expression of the surrounding genes (Donsante et al., 2007; Wang et al., 2012). While those findings have recently been reproduced in a large study using various different serotypes and longer time points (up to 24 months) (Chandler et al., 2015), two additional reports eventually could not reproduce those findings (Bell et al., 2005; Li et al., 2011), but the developmental timing of injections was different, suggesting age-difference susceptibility to rAAV genotoxicity. Additionally, a detailed genomic analysis demonstrated that the vector integration sites were mapped in

Mir341 (within the *Rian* locus), which only exists in rodents (Chandler et al., 2015). In a select number of subjects with HCC that were not AAV gene therapy recipients, genomic analysis of HCC biopsies identified AAV2 genomes to be integrated in oncogenic driver genes (Nault et al., 2015). Whether or not the reported pathogenicity of wild-type AAV2 is relevant to recombinant AAV use in the clinics is unclear (Gil-Farina et al., 2016). In the specific case of gene transfer applied to neurological disorders, insertional mutagenesis may only be a concern when high doses of vector are injected systemically. It also implies that local injections of small amounts of AAV in the parenchyma or CSF will be safe, even though one paper reported a possible integration of AAV in non-dividing cells such as neurons (Wu et al., 1998).

Clinical Progress: Illustrative Case Studies

Clinical gene therapy has come a long way to overcome perceived and real concerns about safety and its potential to markedly intervene in disease in humans, and today the demonstrated proof of concept of the clinical utility of gene therapy is clear. Remarkably, two of those pioneering programs are in nervous system disorders and both lean on AAV. The FDA in 2017 and the European Medicines Agency (EMA) in 2018 approved for the market Luxturna (voretigene neparvovec-rzyl), a subretinally injected AAV encoding the cDNA for RPE65 for treatment of patients with confirmed biallelic RPE65 mutation-associated retinal dystrophy that leads to vision loss and may cause complete blindness in certain patients (Russell et al., 2017). In late 2018, an application was submitted to the FDA, the EMA, as well as Japanese regulators for the market approval of AVXS-101 or Zolgensma (onasemnogene abeparvovec), the Novartis/AveXis AAV gene therapy for SMA (Mendell et al., 2017). If approved—at the earliest in the first half of 2019—it would become the second AAV product currently on the market.

Here, we present several case studies of progress toward or in the clinic of AAV gene therapies targeting the nervous system, with a specific emphasis on diseases of the CNS (summarized in Table 1). For an in-depth overview of the progress and clinical studies in the field of ophthalmology, specifically for inherited retinal disorders, we refer to a number of recent reviews on the subject (DiCarlo et al., 2018; Trapani and Auricchio, 2018). Only a few clinical studies have explored AAV gene transfer applied to inherited peripheral neuropathies, with only one phase I/IIa clinical trial registered so far for the treatment of Charcot-Marie-Tooth disease (NCT03520751) and of giant axonal neuropathy (that affects both PNS and CNS, NCT02362438), yet without any reported results.

Neurosensory Disorders

In some ways, it is remarkable that the first convincing demonstration of the clinical impact of *in vivo* gene therapy was one in a neurosensory disorder; in 2008, the now FDA-approved Luxturna was shown to benefit patients suffering from a type of Leber congenital amaurosis due to mutations in the gene RPE65 (Maguire et al., 2008). This was remarkable, because, historically, gene therapy was primarily considered for disorders with significant mortality, such as muscular dystrophy or genetic forms of immunodeficiency (Miller, 1992). In hindsight, the rationale for the pursuit for therapeutic gene transfer in neurosensory

disorders appears obvious: the tissue target is compartmentalized, warranting a local injection that can be positioned in close proximity to the target cells and thereby limits systemic exposure, a smaller dose requirement, and lessened safety concerns. Approaches in the retina have been on the forefront for the past two decades, resulting in Luxterna and multiple other gene therapy programs for blinding disorders currently in the clinic (reviewed by DiCarlo et al., 2018; Trapani and Auricchio, 2018). The majority of the clinical programs have been based on AAV due to its high gene transfer efficiency to retinal pigment epithelial cells and photoreceptors following a subretinal injection and the availability of many AAV variants that demonstrate altered specificity for the various retinal cell types (Vandenbergh and Auricchio, 2012). The success and growing level of comfort around the use of AAV in the retina continues to push boundaries with the first *in vivo* experimental clinical trials using optogenetics and gene-editing techniques, such as CRISPR, to be targeting the retina and building on the safety and potency of AAV (DiCarlo et al., 2018).

An emerging field that draws on analogies with retinal gene therapy is focused on therapeutic gene transfer for hearing and balance disorders (Géléoc and Holt, 2014). Indeed, the cochlea is also a well-confined space allowing for localized infusions at low doses. Several important distinctions, however, may explain the fact that hearing gene therapy has not seen to date the clinical advances seen in ophthalmology: only recently have novel AAV technologies emerged that improve targeting efficiencies (György et al., 2017, 2018; Landegger et al., 2017); drug delivery into the inner ear is not routine; hearing aids and cochlear implants are a standard of care that can provide therapeutic relief to patients; and lastly, many inherited, particularly autosomal recessive forms, hearing loss disorders likely have either no or only a very short therapeutic window of intervention, thus complicating the translation of gene therapies for these disorders. Nonetheless, the large remaining unmet need and the increasing comfort around AAV gene therapy development is making this field chart ahead, with some notable preclinical advances in genetic forms of hearing loss through gene augmentation (Akil et al., 2012; Dulon et al., 2018; Isgrig et al., 2017; Pan et al., 2017), gene silencing (Shibata et al., 2016), and optogenetic approaches to optimize the resolution of cochlear implants (Keppeler et al., 2018; Wrobel et al., 2018).

Gene therapy research in promising areas, such as olfaction and pain, is growing and in many cases also explores the utility of AAV, most often using local routes of administration (Guedon et al., 2015; Williams et al., 2017).

Lysosomal Storage Diseases

Biochemically, LSDs are characterized by the abnormal degradation of the lysosomal content and the pathological accumulation of undigested materials that leads to cellular damage and multi-systemic clinical manifestations. Most LSDs have a progressive neurodegenerative course that cannot be improved by enzyme replacement therapy as most enzymes do not cross the BBB (Platt et al., 2018), even though allogeneic hematopoietic stem cell transplantation can provide some therapeutic benefit for patients. While those disorders taken individually are rare, the incidence of LSDs as a group is estimated as 1 in 5,000. Gene therapy applied to the treatment of LSDs is especially

attractive for several reasons: (1) LSDs are monogenic diseases (recessive or X-linked), (2) the natural history of those disorders is well established, (3) there is a therapeutic window during which an intervention can take place before overt neurodegeneration, (4) mutation carriers can easily be identified from families, (5) even low levels of residual enzymatic activity can result in significant clinical improvement, and (6) those diseases are often characterized by a severe phenotype and no alternative therapeutic options are available to correct the neurological deficits in patients. In addition, cross correction is made possible due to the fact that the lysosomal enzymes can be secreted in the extracellular space and taken up by surrounding cells, amplifying the corrective effect of the therapeutic gene.

Mucopolysaccharidoses (MPSs) encompass a group of rare metabolic diseases that arise from a deficit in the degradation of mucopolysaccharides and the resulting accumulation of glycosaminoglycans in the lysosome. The age of clinical onset directly correlates with the residual enzyme activity, which is why symptoms can be detected from the prenatal period through adulthood. CNS involvement encompasses difficulty of speech, ataxia, weakness, and dyskinesia in adults, while neurological impairment, developmental delay, and regression are observed in children whose life expectancy is significantly shortened (Platt et al., 2018). Several gene therapy clinical trials are currently in progress for the treatment of MPS I (Hurler infantile syndrome or Hurler-Scheie and Scheie for the juvenile and adult forms), MPS II (Hunter syndrome), MPS IIIA and MPS IIIB (Sanfilippo syndromes A and B), and MPS VI (Maroteaux-Lamy syndrome, which will not be discussed here as the central involvement is not a primary clinical symptom).

The MPS I (NCT02702115) and II (NCT03041324) phase I/II trials are of particular interest as they will both evaluate the safety and tolerability of ascending doses of rAAV2/6-based ZFN therapeutic agents to insert a correct functional copy of alpha-L-iduronidase (IDUA) or iduronate 2-sulfatase (IDS), respectively, into the albumin locus of patient hepatocytes after intravenous delivery. For those studies, adult MPS I and MPS II patients are being recruited, and it is expected that the benefit will mostly be on the peripheral symptoms of these diseases. Whether or not a similar approach can be proposed for younger patients remains to be seen. This would be of interest considering that ZFN-mediated genome editing also prevented the development of neurocognitive deficits in young MPS II mice, even after systemic administration (Laoharawee et al., 2018).

Because MPS IIIA is predominantly a CNS disorder, the phase II/III gene therapy proposed for this disease ("AAVance") will directly target the CNS via stereotaxic infusion of AAVrh10-h.SGSH. The main goal is to deliver a functional copy of the human N-sulfoglucosamine sulfohydrolase (h.SGSH) cDNA in both hemispheres of the brain through MRI-guided neurosurgery. This study will enroll 20 patients of 6 months and older. A phase I/II clinical trial (NCT01474343) by the same team had been previously done on four young patients (6 months to 2 years) with the aim of expressing altogether the human SGSH and SUMF1 (sulfatase-modifying factor, a catalytic activator of SGSH) cDNA after intracerebral administration of AAVrh.10. Overall safe, there was only an indication of cognitive benefit in the youngest subject, suggesting that the disease may have been too advanced in

other patients for a measurable improvement (Tardieu et al., 2014). A parallel approach by intravenous delivery of AAV9.hSGSH is being developed by Abenova Therapeutics (phase I/II ABO-102 trial, NCT02716246). Early-phase clinical trials for the treatment of MPS IIIB (NCT03300453 and NCT03315182) propose to restore expression of N-sulfoglucosamine sulfohydrolase (NAGLU) in patients after intracerebral or intravenous delivery of AAV5 and AAV9, respectively. Both strategies have showed positive impact on the clinical symptoms in rodent and dog models (Ellinwood et al., 2011; Fu et al., 2011).

Neural ceroid lipofuscinoses or Batten disease are a group of neurodegenerative storage disorders. One type of Batten—late infantile neuronal ceroid lipofuscinosis (LINCL)—is a fatal childhood neurodegenerative LSD caused by mutation in the CLN2 gene encoding for the tripeptidyl peptidase-I (TPP-I) protein. The CNS involvement for this disease is significant (myoclonic epilepsy, loss of cognitive and motor function, degeneration of the retina leading to blindness, and early death; Donsante and Boulis, 2018). Currently, four AAV-based clinical trials for Batten disease are under way (three are active and not recruiting while one is recruiting). Pre-clinical safety and efficacy of AAV-based gene therapy for Batten disease has been demonstrated in rats and African green monkeys (Sondhi et al., 2012). The first phase I trial (NCT00151216) started in 2005 with a protocol of intracerebral injection of AAV2CUhCLN2. Ten CLN2 disease patients, with the severe (group A) or moderate (group B) form of the disease, were enrolled in the study. Unfortunately, one of the patients deceased 49 days after treatment from uncontrollable status epilepticus. While imaging studies did not detect a significant change in disease progression, neuropsychological evaluation demonstrated some retardation of disease progression relative to historic controls. It is possible that the use of AAV2, which spreads less than many newly characterized serotypes, did not lead to enough production of TPP-I *in situ*, therefore leading to little therapeutic impact (Worgall et al., 2008). A follow-up phase I study conducted by Weill Cornell (NCT01161576) administered an AAVRh.10CUhCLN2 by direct intraparenchymal stereotaxic injection. In this study, abnormal T2 hyperintensities detected by MRI around the sites of injection led to lowering the dose for the remaining subjects to be enrolled in protocol. Another parallel study with enlarged inclusion criteria (broader genotypes are being included) is being conducted by the same institution on a smaller number of patients (NCT01414985).

Hereditary Leukodystrophy: Canavan Disease

Canavan disease (CD) is an autosomal recessive leukodystrophies caused by loss-of-function mutations in the gene aspartoacylase (ASPA). ASPA is broadly expressed in various tissues but plays an essential role in the degradation of N-acetylaspargate (NAA), one of the most abundant metabolites in the brain (Rigotti et al., 2011; Tallan, 1957). If deficient, NAA abnormally builds up in the CNS, leading to NAA acidemia and aciduria as well as a broad range of deleterious effects such as dysmyelination and spongiform degeneration in the CNS (Appu et al., 2017; Janson et al., 2006). During life, patients commonly present with head lag, macrocephaly, hypotonia, ataxia, inadequate visual tracking, epilepsy, and intellectual disabilities (Hoshino and Kubota, 2014), and many patients die before adolescence.

Preclinical studies in ASPA knockout rodent models demonstrated the remarkable efficacy of AAV expressing ASPA to resolve most of the biochemical and neuropathological hallmarks of the disease (Matalon et al., 2003; McPhee et al., 2005). The strong rationale for ASPA gene therapy has logically led to a first phase I/II dose-escalating clinical trial for CD in 2012 (Leone et al., 2012). Three groups of patients (of 3 months old to 8 years old) underwent AAV2.ASPA stereotaxic injection into the parenchyma at six sites. A control group did not receive any intervention. Adverse events within the first 90 days after treatment were related to the neurosurgery procedure and generally transient. Immune responses to AAV2 appeared limited. NAA concentrations were measured longitudinally by single-voxel ¹H-MRS and showed a monotonic increase of NAA in all regions of interest in the brain of control subjects. By contrast, the pathological accumulation of NAA was decreased in AAV.ASPA-treated patients, which was significant in the periventricular and frontal regions (primary targets for the therapy). Even more interestingly, some patients enrolled in this study showed an arrest or even reversal of brain atrophy assessed by MRI. This was not the case for other subjects who already had a significant degree of atrophy before gene therapy, which makes the case for intervening as early as possible for this disease. Finally, modest but significant improvement in the motor functions (for example, rolling ability) and other clinical outcomes (in particular the level of alertness) was observed in AAV.ASPA-treated patients at 18 months, which was again less pronounced in the older subjects. Decreased clinical seizure frequency was observed in 11 of 13 patients. Overall, the CD clinical trial confirms the long-term safety of intracerebral stereotaxic infusion of AAV and shows evidence of positive radiographic and clinical changes, which is not 10 years out for some patients.

Neurotransmitter Disorder: Aromatic L-amino Acid Decarboxylase Deficiency

Aromatic L-amino acid decarboxylase (AADC) is a key enzyme involved in the metabolic biosynthesis of dopamine and serotonin, two monoamine neurotransmitters. The clinical manifestation of AADC deficiency includes hypotonia, hypokinesia, dystonia, oculogyric crisis, and autonomic dysfunction (Hwu et al., 2013). In the most severe form of the disease, children never reached developmental milestones and are unable to hold their head, sit, or stand. Life expectancy for those patients does not go above 5–6 years of age, as no cure is available. The first gene therapy trial for AADC deficiency (Hwu et al., 2012) was inspired by other clinical studies using AAV-based gene transfer in PD delivered into the putamen, which also aimed at restoring AADC expression and was well tolerated (Christine et al., 2009; Muramatsu et al., 2010). This first trial included four pediatric patients who received CT-guided stereotaxic injection of AAV2.CMV.hAADC in the putamen and whose motor performances were compared with untreated patients in the natural history control group. Impressively, the treated subjects gained weight, and their motor scores and cognitive functions gradually improved after gene transfer, even though all of them already had substantial impairment before intervention. All of them achieved head control and could sit with support. Results from a follow-up study including ten more subjects have been recently published, confirming the earlier findings (Chien et al., 2017).

(NCT01395641). Currently, three clinical trials following a similar protocol (intra-putamen injection of AAV2.hAADC) are ongoing (NCT01395641, NCT02852213, and NCT02926066), with two of them actively recruiting new patients.

Idiopathic Age-Related Neurodegenerative Diseases

Gene therapy, conceptually straightforward for monogenic disorders, is more challenging for sporadic age-related neurodegenerative diseases. Because those diseases often result from the influence of many different pathogenic alleles with small individual impact, finding a good therapeutic target can be challenging despite the well-characterized neuropathological hallmarks. In addition, the lack of a clear genetic marker to identify patients and the relative frequency of mixed pathology often hindered the recruitment of a homogeneous set of patients in clinical trials. On the other hand, the high incidence of those idiopathic diseases facilitates the design of double-blind, randomized, and placebo-controlled studies including larger groups of patients (up to 60 for NCT00985517, a gene therapy trial for PD). Of note, in these types of studies, subjects in the placebo group are subjected to a sham injection protocol that does not include a brain cannula, injection, or vector infusion in order not to subject the individual to the risks associated with the procedure.

PD is a common neurodegenerative disorder characterized by bradykinesia, rigidity, tremor, and gait dysfunction. The main neuropathological hallmark of this disease is a loss of dopaminergic neurons in the substantia nigra pars compacta (SNc). The first line of defense to compensate for the loss of dopamine from the SNc is for patients to take L-dopa, which will be actively converted to dopamine by AADC. However, as the dopaminergic neurons continue degenerating, the effect of L-dopa weans off with time. Logically, one of the most straightforward application of gene therapy for PD is to restore expression of AADC by AAV transduction of non-degenerating striatal neurons (the site of action of dopamine). Pre-clinical results have proven that this strategy could increase the level of dopamine over a long period of time (up to 7 years in NHPs) (Bankiewicz et al., 2000; Bankiewicz et al., 2006; Sánchez-Pernaute et al., 2001), and the results of one phase I safety study have been published (NCT00229736) (Christine et al., 2009; Eberling et al., 2008). A total of ten patients were initially injected in two dose cohorts with AAV2.CMV.hAADC vector targeting four sites in the putamen. The intervention appeared to be well tolerated, even though postoperative MRI scan revealed the presence of hemorrhagic events along the cannula trajectory. Nevertheless, improvement of the Unified Parkinson's Disease Rating Scale (UPDRS) scores was reported at 6 months, which persisted for at least 2 years. Those remarkable results logically prompted the launch of other similar trials (either active or currently recruiting) with higher doses of vector (NCT02418598, NCT01973543, and NCT03065192) and larger cohorts of patients (up to 42 individuals for NCT03562494). The first randomized, placebo surgery controlled and double-blinded phase 2 clinical trial for AAV.hAADC (VY-AADC02, Voyager Therapeutics) in advanced PD started in June 2018 (NCT03562494) (Table 1).

Alternatively, AAV-driven expression of neurturin is the focus of three other gene therapy trials for PD. Neurturin is a neurotrophic factor analog of glial-cell-derived neurotrophic factor

(GDNF) that had been shown to improve dopaminergic activity in aged monkeys and be beneficial to animal models of PD (Gasmi et al., 2007; Herzog et al., 2007; Kordower et al., 2006; Kotzbauer et al., 1996). In this case, the therapeutic agent does not directly target a causative pathological molecular pathway but offers neurotrophic support to the vulnerable neuronal population degenerating in the disease. After having demonstrated the safety of neurturin gene transfer in a phase I study (NCT00252850) (Marks et al., 2008), results of the first double-blind, randomized, and sham-surgery-controlled gene therapy trial has been published in 2010 (NCT00400634) (Marks et al., 2010). Patients with idiopathic PD were randomly assigned (2:1) to receive bilateral stereotactic intraputamenal injection of AAV2.neurturin or sham procedure. Except for the neurosurgical team, patients and medical personnel were blind to the nature of the treatment. Assessment of motor functions at 18, but not at 12, months (UPDRS scale) did show some benefit of the AAV.neurturin over the sham-treated group. Post-mortem examination of the brains of two AAV.neurturin-treated individuals (Bartus et al., 2011) revealed that NRTN immunostaining covered, on average, 15% of the putamen, with sparse tyrosine hydroxylase induction in the striatum (no impact on the SNc was observed). Follow-up clinical trials include an open-labeled, one-arm study using convection-enhanced delivery of AAV2.GDNF (NCT01621581) (Johnston et al., 2009; Su et al., 2009) and a second trial double-blind, randomized, placebo-controlled trial at higher dose (NCT00985517).

Late-onset AD is the lead cause of age-related dementia and affects more than 40% of individuals 85 years and older in Europe and the USA (Brookmeyer et al., 2007; Burns and Iliffe, 2009), and no less than 112 therapeutic agents are currently being tested in clinical trials (Cummings et al., 2018). After a first asymptomatic phase (during which amyloid accumulation is already detectable), cognitive complaints are reported by patients, followed by progressive loss of short-term memory, gradual deterioration of cognition, and loss of autonomy. From a neuropathological point of view, amyloid beta peptides aggregate and form extracellular neuritic plaques while the accumulation of abnormally phosphorylated tau protein triggers the development of intraneuronal neurofibrillary tangles. So far, none of the clinical trials have proven to significantly affect the course of the disease (with the exception of the phase II BAN2401 immunotherapy trial, NCT01767311, which may have a modest positive impact on the progression of cognitive deficits in AD patients), and only one attempt at gene therapy has been translated in patients.

This phase I trial, completed in 2014, was aimed at increasing the levels of nerve growth factor (NGF) in the brain of patients (NCT00087789) (Rafii et al., 2014). Similar to the neurturin approach for PD, this therapeutic agent essentially provides neurotrophic support. The rationale for choosing NGF was to prevent dysfunctional changes of basal cholinergic neurons, which are especially vulnerable to AD-associated stressors (Bartus et al., 1982; Hampel et al., 2018; Whitehouse et al., 1982). This approach by AAV-driven gene transfer (AAV2.NGF or CERE-110) also proposed an alternative to using cholinesterase inhibitors, which are associated with marginal improvement but non-negligible side effects. AAV.NGF was injected to the nucleus

basalis of Meynert, the main source of cholinergic input of the neocortex, in ten patients with mild to moderate AD. With the exception of an event of postsurgical hygroma, the surgical procedure was well tolerated. At the 24-month time point, the rate of clinical deterioration of AAV-treated patients did not appear to be exacerbated as compared to the expected rate of decline determined from observational studies. However, potential efficacy could not be tested because of the small size of the cohort and the probable moderate impact of the treatment, even though a marginal stabilization in brain glucose metabolism was observed. Eventually, five patients died from the disease 3.3 to 6 years after treatment, and the post-mortem analysis of their brains showed evidence of NGF expression limited to the area in close proximity to the needle tract. The good safety and tolerability of this therapeutic agent has led to the design of a multicenter, double-blind, sham-surgery-controlled phase 2 clinical trial of AAV2.NGF in patients with mild to moderate AD started in 2009 (NCT00876863). As of July 2018 (24 months post AAV injection), no significant difference was detected between the treatment and placebo groups on clinical outcome measures or on selected AD biomarkers (Rafii et al., 2018). Ways for improvement include the recruitment of patients at an earlier stage of the disease (an MMSE between 17 to 26 is considered as “mild dementia”) and the use of a more efficient vector serotype able to spread in larger area of the brain.

In July 2018, another pending phase I AAV-based clinical trial for AD seeks to assess the safety and toxicity of intracisternal delivery of AAVrh.10hAPOE2 in 15 patients homozygous for the epsilon-4 allele of the apolipoprotein E gene (APOE4/APOE4) (Rosenberg et al., 2018). APOE4 is by far the most significant genetic risk factor of sporadic AD (Corder et al., 1993; Strittmatter and Roses, 1996), while the epsilon-2 allele of APOE lowers this risk by half and even alleviates the severity of familial cases of AD (Corder et al., 1994; West et al., 1994). Because APOE is a secreted protein, significant amounts were also detected in the CSF, therefore possibly affecting a larger portion of the cerebral tissue than the areas primarily targeted by the injections. Whether or not those results will pan out in human AD patients and whether expression of APOE2 in neuronal cells may cause unexpected adverse events will be addressed in the phase I study.

Spinal Muscular Atrophy

SMA is a monogenic disorder caused by the loss of function of the gene encoding the survival motor neuron 1 (SMN1) protein, which leads to the progressive loss of lower MNs. SMA1 is the most severe form of the disease and a common genetic cause of death among infants, with a median age at symptom onset of 1.2 months and a very low rate of survival passed above 20 months (Finkel et al., 2014). Gene therapy application for type I SMA has recently received much attention from the scientific community for several reasons: first, it was the first time that a therapeutic gene was delivered via intravenous injection of an AAV9 harboring a scAAV vector genome (Figure 1C); second, the clinical benefit observed in patients was remarkable (NCT02122952) (Mendell et al., 2017). In the SMN1 gene augmentation approach, 15 infants received a high dose of scAAV9.CB.hSMN (AVX-101) intravenously (Figure 3). The study design included two dose cohorts. The high dose led to in-

creases in serum aminotransferase, a sign of liver toxicity, in the first treated patient, which was managed by oral prednisolone. Subsequent subjects all received prophylactic prednisolone 24 h before surgery and continued for a month after the vector infusion. As of August 2017, all the children had survived and reached 20 months without the need for permanent mechanical ventilation. All subjects presented increased motor functions from baseline (based on the Children’s Hospital of Philadelphia Infant Test of Neuromuscular Disorders, or CHOP INTEND, scale), with greater improvements observed in the high-dose cohort who also had been treated at younger ages. Most of the children of this group achieved head control, sitting, rolling over, or speaking, milestones that otherwise would have never been reached according to the well-documented natural history for this disease (Finkel et al., 2014). This program subsequently was taken to phase 3 studies for SMA type I and follow on clinical studies for other types of SMA (NCT03306277, NCT03505099, NCT03461289, and NCT03381729), results of which are pending (Table 1). Originally developed by AveXis for SMA type I, a Biological License Application was submitted to the US FDA in November 2018 by Novartis, which acquired AveXis.

Conclusions

Alleviating symptoms long term or addressing the underlying etiology of disease is the goal in any area of medicine; however, for disorders of the nervous system, this often becomes a necessity. The limited access to therapeutic tissues like the brain, spinal cord, retina, or cochlea, either pharmacologically or surgically, begs for a lasting, if not permanent, approach to intervene in the disease process. Conceptually, gene therapy can achieve this, by either adding or eliminating a major genetic determinant in genetically defined disorders or, in common complex disease, by introducing a disease modifier in a lasting manner within the difficult-to-reach cell, tissue, or compartment. Over the past decades, increased insights in the challenges to clinical gene therapy, technological improvements that surmount these hurdles, and the advances in disease genetics, mechanism, diagnosis, and management have led to a series of milestones in clinical gene therapy. A growing body of data now supports the safety and, in an increasing number of cases, the efficacy of gene therapies across various modalities of genetic cargo, routes of administration, and target tissues. A common denominator in many of these studies is the use of AAV, a small viral gene transfer vector system with strong neuronal tropism and a desirable safety profile. The utility of AAV has been proven in dozens of clinical studies ranging from small-dose administrations for neurosensory disorders to large-dose systemic injections that achieve broad transduction of MNs in the spinal cord resulting in demonstrable and—at times—remarkable clinical benefit to patients.

While the opportunity is great for bringing gene therapy to inherited rare diseases and beyond, emerging from the clinic remains a challenge even after the compelling proof-of-concept studies. From the scientific perspective, particularly for common complex disorders, gene therapy, like any other drug modality, requires more viable molecular disease targets and modifiers. One prospect is that non-AAV biologics, which in their current

form require repeat administration (e.g., antisense oligonucleotides or antibody-based treatments in IRD or AD), are “de-risked” independently to then be incorporated for sustained delivery in an AAV formulation. From the technological perspective, notwithstanding the progress made, there is a need for further advances such as improved control over the specificity, kinetics, and potency of gene transfer and expression that limit toxicities, unlock new therapeutic target cells such as microglia, and improve on the efficacies already noted. The efficiency of AAV9 for CNS targeting, for example, even with scAAV technology, remains far from saturating and with limited neuronal targeting. Vector engineering efforts highlight the potential of AAV to do better in these and other respects. Particularly for the high-dose systemic AAV applications, improvements are needed to maintain a healthy margin on a therapeutic window. While that window is primarily determined by a toxicity ceiling, in the context of gene therapy currently, manufacturability and cost can also be determinants. The cost associated with the production at scale of the doses needed to treat a single subject, let alone a broader patient population, can be staggering and in some scenario’s cost prohibitive for an otherwise viable gene therapy to move to clinical studies. Lastly, the ability to address the etiology of disease and the potential to induce a durable correction—the properties that make gene therapy so compelling, as Keeler noted in 1947 (Keeler, 1947)—are disruptive to traditional models in the pharmaceutical industry, creating uncertainty to a broader adoption of AAV gene therapy by clinicians, drug makers, and insurers.

Nonetheless, the clinical success in genetically defined diseases, such as IRD and SMA, has catalyzed an increasing activity and creativity to think through these hurdles. The future for AAV gene therapy to make a marked impact for patients suffering from disorders of the nervous system is therefore bright.

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DECLARATION OF INTERESTS

E.H. is a consultant to AZTherapies. L.H.V. holds equity in and chairs the Scientific Advisory Board of Akouos, a gene therapy company focused on hearing disorders. L.H.V. is also inventor and has a royalty interest in various AAV technologies, including AAV9 and AncAAVs, which have been licensed to several of the companies discussed in this Review. L.H.V. is a consultant to Nightstar Therapeutics, Cobalt, Lonza Houston, Exonics, and Selecta Biosciences and has received travel reimbursement from AveXis (now Novartis). L.H.V. receives research funding from Lonza Houston, Oxford Biomedica, Selecta Biosciences, and Solid Biosciences.

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Therapeutic AAV Gene Transfer to the Nervous System: A Clinical Reality

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In the original publication of this review, one of the vector doses listed in Table 1 contained an error. In the Lysogene clinical trial with AAVrh10-hSGSH (LYS-SAF302) for MPS IIIA, a dose of 7.2×10^{12} vg was administered intracranially via 6 injection sites and not a dose of 7.2×10^{11} over 12 sites, as initially stated in the review. This has since been corrected online. The authors apologize for the error.

